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THE ATMOSPHERE

THE ATMOSPHERE IS THE protective blanket that makes life possible on Earth. It is the air we breathe; it protects us and all other organisms that live on the surface from the Sun's deadly ultraviolet (UV) radiation; it insulates Earth and regulates its surface temperature so that liquid water exists now and apparently has always existed somewhere on the surface since nearly the beginning of Earth history. Without the atmosphere, Earth's surface would be frozen, and life would not exist here.

Earth's atmosphere reaches about 500 kilometers (300 miles) above its ocean and land surface. It is layered in terms of composition, temperature, and dynamical behavior, and it has a composition that is unique among planets in our solar system and, if not unique, apparently extremely rare among planets in other solar systems. The atmosphere itself transports the majority of energy from the equator to the poles, and its winds propel ocean-surface currents that comprise the second important part of poleward heat transport. Furthermore, winds transport water evaporated from the ocean to the land, where it may fall as rain or snow, thus driving part of the hydrological cycle. The atmosphere is also one path by which carbon moves between the ocean and the land and other carbon reservoirs (chapter 4).

Despite its importance, the atmosphere constitutes only a minuscule part of Earth (Back-of-the-Envelope Calculation). The total mass of the atmosphere is 5.14×10^{18} kg, which is a mere four-thousandth of the mass of the ocean (1.39×10^{21} kg) and one-millionth that of the solid Earth (5.98×10^{24} kg). If not from the surface, then at least from space, the true extent of Earth's thin skin of atmosphere is apparent (see Prologue opening figure).

This chapter provides a general overview of the nature of the atmosphere and the way it circulates. The important concept of radiation balance and the role of greenhouse gases in climate and climate change are discussed separately and in detail in chapters 5 and 6.

The Composition of Air

From a planetary perspective, Earth's atmosphere is unique because it contains oxygen (O_2). No other planets in our solar system, and probably very few in other solar systems, possess oxygenated atmospheres. In fact, early in its history, Earth's atmosphere was also essentially



BACK-OF-THE-ENVELOPE CALCULATION

The Vastness of the Atmosphere

The atmosphere is very important for life on Earth and moves a tremendous amount of energy and mass around the surface of the planet. Despite its great importance, however, the atmosphere represents a very small portion of Earth's thickness and mass. To help put this in perspective, let's consider a few scaling exercises to help place the relative quantities in context.

The diameter of the solid Earth is about 12,742 kilometers (7,918 miles), while about 80 percent of Earth's atmospheric mass is in the troposphere, with an approximate height of 16 kilometers (52,000 feet) above Earth's surface. Those are distances that are hard to grasp, so let's consider scaling the numbers to something more familiar. The diameter of a men's professional basketball is about 25 centimeters (9.8 inches). If we represent Earth with the basketball, how thick would its corresponding atmosphere be? The answer is simple: we just have to multiply 25 centimeters by the ratio between the thickness of the troposphere and the diameter of Earth (0.12 percent). The answer is 0.03 centimeter (0.01 inch), or about the thickness of three sheets of standard paper.

We can perform a similar calculation for mass. The mass of Earth is about 6.0×10^{24} kg, while the mass of the troposphere is about 75 percent of the mass of the atmosphere, or 3.85×10^{18} kg. The mass ratio of the troposphere to the solid Earth is therefore about 6.4×10^{-7} . Because this ratio is much smaller than the distance ratio, let's consider a more massive object that is familiar to us all: an automobile. Let's assume that the average mass of a car is about 1,500 kilograms (3,307 pounds). If the solid Earth were only as massive as a car, the proportional mass of the atmosphere would therefore be about 1 gram (0.035 ounce). That is the approximate mass of a paperclip!

Both of these exercises are surprising, given our everyday experience with the vastness of the atmosphere. Despite its apparent scale from our perspective, however, the atmosphere is a very thin veneer that encapsulates our vast planet.

O₂-free. The early atmosphere may have been dominated instead by nitrogen-bearing compounds. Nitrogen was part of the material that accreted to form Earth, but because none of the minerals that solidified from that early molten mass take up nitrogen, it was likely left as a major component of the early atmosphere. The addition of oxygen to the atmosphere apparently began later, with the emergence of photosynthesis as an important metabolic process. By 2.4 billion years ago—more than 2 billion years after Earth formed—an atmosphere containing perhaps a few tenths of a percent of O₂ began to develop. The reasons, therefore, that Earth's atmosphere is unique among planets are that Earth harbors life and that early Earth provided an environment conducive not just to the establishment of life, but also to the evolution of complex forms of life.¹

For the next 2 billion years, Earth's atmosphere probably contained only a few tenths to thousandths of a percent of O₂, with the modern O₂-rich atmosphere developing just several hundred million years ago.² As O₂ built up in the atmosphere, so did ozone (O₃). Ozone is notable because it absorbs UV radiation, which is lethal to life in large doses. According to

the fossil record, life spread from water to land about 400 million years ago. Thus the emergence of terrestrial life may have been made possible by the presence of enough O_3 in the upper atmosphere to shade the surface from UV radiation.

Today, just three gases—nitrogen (N_2 [78.1 percent]), O_2 (20.9 percent), and argon (Ar [0.93 percent])—make up 99.96 percent of dry air by volume. The remaining gases are carbon dioxide (CO_2), methane (CH_4), O_3 , nitrous oxide (N_2O), and a host of other “trace” gases—those that exist in only minuscule quantities. Additionally, water vapor is almost always present in the atmosphere but in varying amounts. Its concentration depends strongly on temperature and on a hot day may reach several percent regionally.

In the context of climate, water vapor and CO_2 are particularly important because both are greenhouse gases, but their roles differ in important ways. The *anthropogenic* (due to human activities) rise in the CO_2 content of the atmosphere is the root cause of the current warming, but water vapor acts as a feedback; that is, its concentration increases due to the warming from other causes, and by being a greenhouse gas itself, it amplifies that warming. Carbon dioxide also has a long residence time in the atmosphere of thousands of years, while the water vapor content fluctuates rapidly. At Hawai’i’s Mauna Loa Volcano monitoring station, the average annual concentration of CO_2 in the atmosphere first exceeded 400 parts per million (ppm) by volume (0.0400 percent) in 2015 (chapter 4). This concentration is up from about 280 ppm at the beginning of the Industrial Revolution around 1750, and is currently rising at a rate of 2.7 ppm a year. Other greenhouse gases—such as CH_4 , O_3 , and N_2O —are also on the rise but exist in much lower abundances and have limited residence times in the atmosphere. Some of the trace gases are also potent greenhouse gases. The differing roles of CO_2 and water vapor, as well as the contributions to warming by the other greenhouse gases, are described in detail in chapter 6.

The Compressibility and Pressure of Air

A fundamental property of air—for that matter, any gas—is its compressibility, or the amount that its volume changes in response to the pressure that is exerted on it. The higher the pressure on the gas, the smaller its volume becomes (and the greater its density). The volume also depends on temperature, such that the higher the temperature, the higher the volume (and the lower the density). The temperature (T), pressure (P), and volume (V) of gases (in the ranges that they are found in the atmosphere) are related by the *ideal gas law*,³

$$PV = nRT,$$

where n is the number of moles of gas (1 mole = 6.02×10^{23} molecules [or atoms for monoatomic gases]) and R is the universal gas constant (8.314 J/mole K, where J is a joule, a unit of energy, and K refers to the unit of temperature known as a Kelvin; 1 Kelvin is equivalent in magnitude to 1 degree Celsius, but the two temperature scales are different [273 K is the same temperature as 0°C]).

Because air is compressible, its weight causes air pressure to be much greater at the surface than at altitude. The mass of the atmosphere is such that a column of air 1 centimeter

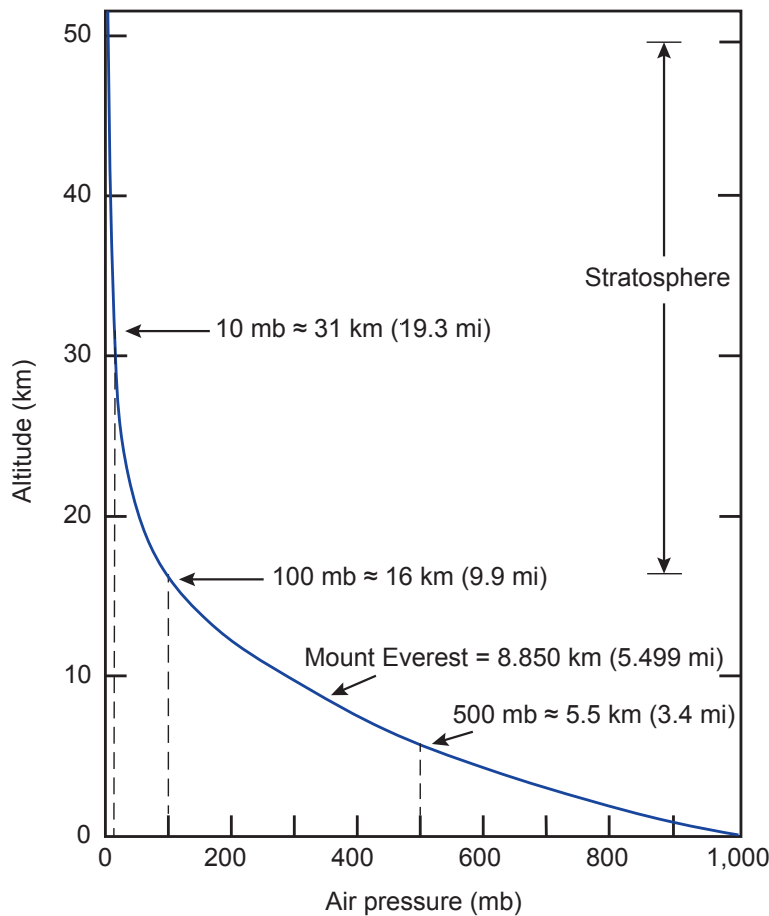


FIGURE 1.1 The decrease in air pressure with altitude

Air pressure is highest at the surface because of the weight of the overlying air, which amounts to $1,034 \text{ g/cm}^2$ (14.7 lb/in^2). Half of the mass of the atmosphere is below an altitude of 5.6 kilometers (3.5 miles).

(0.394 inch) on a side at its base and extending from sea level to the outer limit of the atmosphere weighs $1,034 \text{ grams}$ ($10,340 \text{ kg/m}^2$ [14.7 lb/in^2]). This is the “air pressure” around us. Air pressure decreases exponentially with altitude (figure 1.1). In other words, there is a lot more air in a given volume at Earth’s surface than in the same volume high in the atmosphere, and the air near the surface is relatively dense, whereas air high in the atmosphere is “thin.” For an atmosphere at its mean temperature, about 50 percent of its mass is below an altitude of 5.6 kilometers (18,370 feet) above sea level.

Mechanisms of Heat and Mass Transfer

THE CONVECTIVE MOTIONS OF THE ATMOSPHERE

The relationships among the pressure, temperature, and volume of air help explain why the atmosphere is in continuous convective motion. *Convection* refers to the buoyant rising of

less dense fluid (gas or liquid or, for that matter, even a ductile solid) and the sinking of more dense fluid.⁴ The lower part of the atmosphere is heated primarily from below by Earth's surface. The density of a parcel of air that is warmed at the surface decreases (becomes more buoyant), and that parcel rises if the air around it is cooler and therefore denser. The heating of the air parcel occurs through the transfer of *sensible heat*, or heat associated with a change in temperature, and occurs principally by conduction from Earth's surface. *Conduction* refers to the transfer of heat by collisions between molecules of warmer matter, in which the molecules possess relatively high kinetic energies, and those of cooler matter, in which the molecules possess relatively low kinetic energies. When these collisions occur, they increase the kinetic energy of the molecules in the cooler matter and therefore increase its temperature.

In the atmosphere, convection occurs on all scales from meters (hot air rising from a chimney), to kilometers (warm air rising and cold air descending below a thunderhead), to 1,000 kilometers (620 miles) or more (convective rise of the atmosphere over the warm equatorial Pacific Ocean). On the large scale of weather patterns, convection spawns winds to replace the rising air. This reflects the fact that rising air masses create zones of relatively low atmospheric pressure, and sinking air masses result in zones of relatively high atmospheric pressure. Again, this occurs on various scales and intensities, from the extreme low pressures that characterize the 100 kilometer-wide (62-mile) cores of hurricanes (also termed cyclones and typhoons), to the gentle highs that characterize the global-scale belts of deserts in the subtropics.

ADIABATIC TEMPERATURE CHANGE

Vertical motions of air result in adiabatic (from the Greek word *adiabatos* [impassable]) heating or cooling. An *adiabatic* process is one in which there is no change in the heat content of the system. For the atmosphere, this means that a particular parcel of air that adiabatically rises or sinks neither gains nor loses heat. This requires that the rising or sinking is accompanied by a change in the temperature of the air. For example, as an air parcel rises and expands due to the pressure decrease, its temperature drops. In the idealized case, the temperature reduction is in proportion to the extent of the volume increase. The reason is that the expansion of the parcel of air represents an expenditure of energy on its surroundings, and the temperature drop represents that expenditure. We commonly experience this phenomenon when, after releasing gas from a pressurized can of, say, spray paint or deodorant, we find that the can has cooled. Conversely, if the parcel is adiabatically brought back to its original pressure, its volume decreases and its temperature increases, with both returning to their original values.⁵ For a thermodynamic explanation of adiabatic changes, see box 1.1.

WATER VAPOR IN THE ATMOSPHERE

Water vapor plays an important role in climate because it is a greenhouse gas, serves as an important mechanism of heat transport, and forms clouds and precipitation.⁶ The amount

BOX 1.1

Thermodynamic Background

Adiabatic processes are ones in which no heat is exchanged between a system and its surroundings. To understand adiabatic processes, it is useful to consider the *First Law of Thermodynamics*,

$$\Delta E = E_2 - E_1 = q + w,$$

where ΔE is the total change in energy of the system, E_2 and E_1 are its final and initial states, q is the heat change, and w is the work done on the system. (Here we choose the convention whereby positive values of q and w mean $E_2 > E_1$.) The First Law states that (1) energy is always conserved (it is neither created nor destroyed in any non-nuclear process), and (2) heat and work are equivalent (both are forms of energy transfer from one to another system).

With the First Law in mind, now let us consider what happens to a parcel of air as it rises several kilometers into the atmosphere. To help our thinking, let us imagine the air parcel as a spherical balloon with a mass that remains constant; that is, the walls of the balloon are impermeable (figure B1.1.1). (This is actually not too far from reality. Because the thermal conductivity of air is low and air density depends on temperature, in the real atmosphere air tends to exist and move around as thermally discrete parcels.) As the balloon rises, the atmospheric pressure around it decreases. The volume of the balloon thus increases so that the inside pressure equals the outside pressure. The expansion of the system (the balloon) against its surroundings is the *work* done by the system (and thus has a negative value) and is given by the expression

$$w = -P\Delta V.$$

If the rise is adiabatic, the balloon neither gains nor loses heat, so $q = 0$. Then, according to the First Law, the expenditure of work by the system during the expansion must equal its expenditure of energy:

$$\Delta E = q + w = 0 - P\Delta V.$$

Because the change in internal energy is proportional to the temperature change, we may write,

$$-P\Delta V = \text{constant} \times \Delta T.$$

In other words, $\Delta T \neq 0$; more specifically, T (and ΔE) must decrease in proportion to the work done by the system on its surroundings. Conversely, were the balloon to be brought back to the surface, its volume would decrease and the air pressure and temperature inside it would increase, with all three returning to their original values if the expansion and contraction of the balloon were done reversibly.

The ΔT for a reversible, adiabatic expansion of an ideal gas may be calculated from the heat capacities (the ratio of heat change to temperature change of a substance, discussed further in chapter 2) of the gas at constant V (C_v) and at constant P (C_p), with the assumption

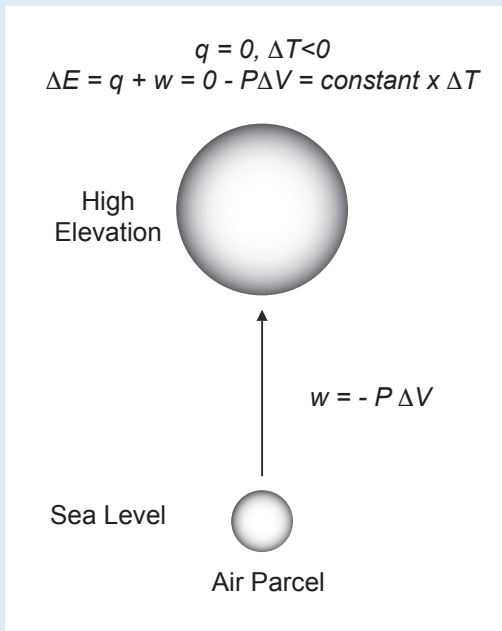


FIGURE B1.1.1

Changes in the temperature of and work expended by a parcel of air of constant mass as it adiabatically rises in the atmosphere.

that C_v remains constant over the temperature interval of the expansion. Taking $\gamma = C_p / C_v = 1.4$ for air, then

$$T_2 / T_1 = (V_1 / V_2)^{\gamma-1}$$

$$T_2 / T_1 = (P_2 / P_1)^{(1/\gamma)},$$

where T_1 , V_1 , and P_1 refer to the initial, known condition and T_2 , V_2 , and P_2 refer the final condition, where one must be specified.

Because the adiabatic rise of air results in cooling, it commonly leads to condensation of water vapor. The phenomenon, as noted, involves the release of latent heat, an important mechanism of heat transfer in the atmosphere. The relationship between the amount of water vapor the atmosphere can hold and its temperature is expressed by the *Clausius-Clapeyron equation*,

$$\ln (P^{H_2O} / P_0^{H_2O}) = (\Delta H_{vap} / R) (1 / T_0 - 1 / T),$$

where $P_0^{H_2O}$ is a known partial pressure of water vapor at a known temperature T_0 (in Kelvins), P^{H_2O} is the partial pressure of water vapor at temperature T to be determined, ΔH_{vap} is the heat (or enthalpy) of vaporization of water (≈ 44.0 kJ/mole at 25°C), and R is the universal gas constant. The Clausius-Clapeyron equation makes several assumptions, the most important of which is that ΔH_{vap} remains constant over the temperature range of the calculation. ΔH_{vap} is never truly constant. In the case of water, for example, ΔH_{vap} decreases from 44.9 kJ / mole at 0°C to 40.6 kJ / mole at 100°C . The Clausius-Clapeyron equation was used to calculate the curve in figure 1.2.

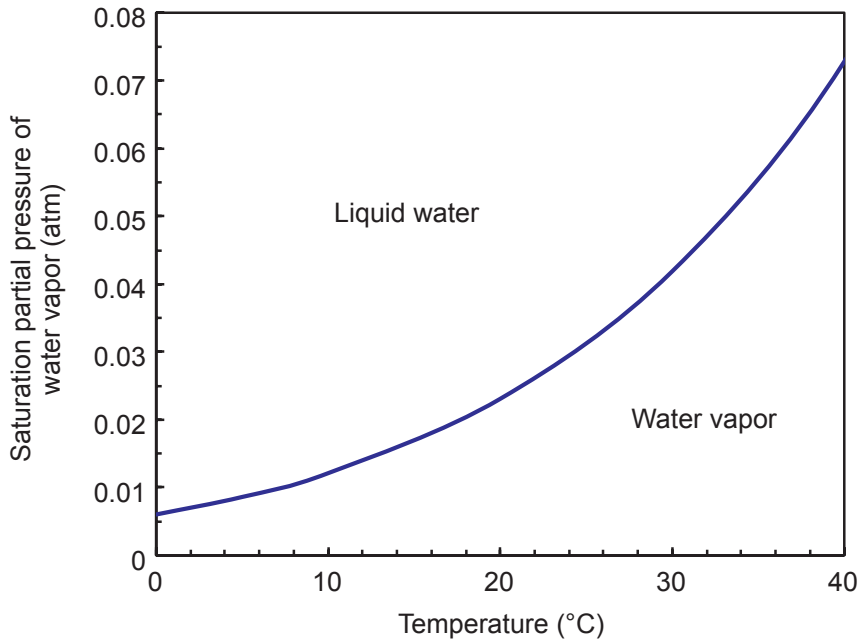


FIGURE 1.2 The water vapor pressure in air in equilibrium with liquid water as a function of temperature at 1 atmosphere total pressure

The curve was calculated from the Clausius-Clapeyron equation.

of water vapor in air is referred to as *humidity*, which is commonly expressed as *relative humidity* (the partial pressure of water vapor in the atmosphere relative to the saturation partial pressure, usually expressed as a percentage) or *specific humidity* (the mass of water vapor per unit mass of dry air, usually given in g/kg). The amount of water vapor that can exist in the atmosphere depends strongly on temperature (figure 1.2).

This brings us back to adiabatic cooling. As a consequence of the dependence of water vapor content on temperature, the water content of the atmosphere drops rapidly with altitude. For example, the specific humidity at an altitude of 5 kilometers (16,400 feet) is only about 15 percent of that at the surface due mainly to the drop in temperature with altitude. For the same reason, the mean annual humidity at Earth's surface dramatically drops from the tropics to the high latitudes (figure 1.3); that is, the cold polar atmosphere is quite dry, whereas the air in the tropics is commonly moist. Similarly, as a parcel of air rises in the atmosphere it first cools, as we described for the dry adiabatic conditions. The temperature of the parcel eventually reaches the *dew point*, the temperature at which the water vapor in the air parcel begins to condense.

Adiabatic cooling and condensation have two important consequences. First, they result in the formation of clouds or precipitation (figure 1.4). Second, the condensation releases what is known as latent heat. *Latent heat* is defined as the energy released or absorbed by a phase change of matter, meaning a change from one state (gas, liquid, solid) to another state, without a change in temperature (see figure 2.2). Thus the evaporation of 1 mole of water requires 44.0 kilojoules (kJ), and the same amount of energy will be released when that 1

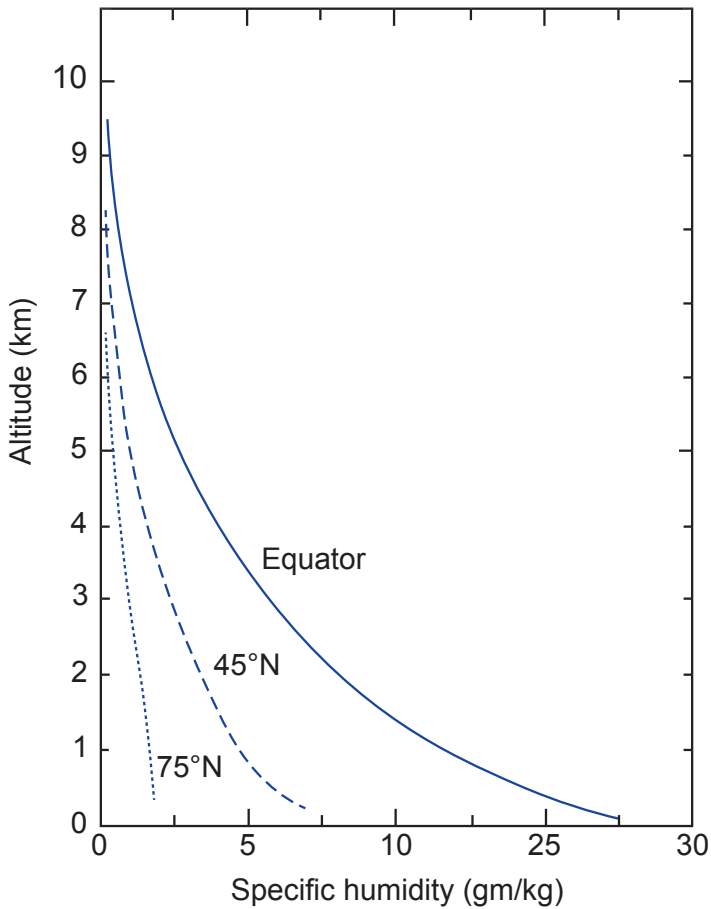


FIGURE 1.3 Profiles of mean annual humidity as a function of altitude at different latitudes

The dramatic drop in humidity with altitude and the overall much lower humidity at high latitudes compared with the tropics reflect the dependence of water vapor pressure on temperature. (Redrawn from D. L. Hartmann, *Global Physical Climatology* [San Diego, Calif.: Academic Press, 1994])

mole of water vapor condenses, with no change in temperature accompanying the evaporation or condensation. The absorption of latent heat by evaporation of water at the surface and the release of that latent heat by its condensation at altitude is an important mechanism of upward heat transfer in the troposphere. In fact, over the ocean and land surfaces combined, latent-heat transfer is nearly five times larger than sensible-heat transfer.

The Thermal and Compositional Layering of the Atmosphere

The physical laws and properties that we have discussed have important implications for the vertical structure of the atmosphere, which, in turn, is important for weather and climate. The atmospheric layer closest to Earth is called the *troposphere* (figure 1.5).⁷ The troposphere accounts for about 80 percent of the total mass of the atmosphere. The atmospheric transport of heat from the tropics to high latitudes occurs almost entirely in the troposphere. It is



FIGURE 1.4 *Cumulonimbus clouds over Africa formed by the convective rise of warm, moist air*

The enormous cloud has risen to the tropopause, where the winds have flattened its top to form a disc-like structure known as an anvil. Due to continued strong convection, the central cloud “tower” can be seen poking up through the center of the anvil. These characteristic vertical cloud “towers” spawn thunderstorms if they contain enough moisture. A crew member of the International Space Station took this photograph. (NASA, International Space Station Imagery, <https://spaceflight.nasa.gov/gallery/images/station/crew-16/html/isso16eo27426.html>)

where the weather occurs and where most of the planet-insulating greenhouse gases are concentrated. The temperature of the troposphere drops rapidly with altitude to reach a minimum at the *tropopause*, which separates the troposphere from the overlying stratosphere. This boundary is at an altitude of 16 to 18 kilometers (52,000–59,000 feet) in the tropics, but drops to 8 to 10 kilometers (26,000–33,000 feet) and becomes more diffuse in the polar regions. The drop is gentle from the equator to about 40°N and 40°S latitudes, but there the tropopause rapidly plunges about 2 kilometers (6,600 feet), owing to the circulation patterns in the troposphere, before continuing its gentle drop to the poles. The height and thickness of the tropopause with latitude mainly reflect the fact that incoming radiation is greater than outgoing radiation in the tropics, while the opposite is true in the high latitudes.

The tropopause exists because of the thermal stability of the overlying *stratosphere*, which extends from the tropopause to an altitude of about 50 kilometers (30 miles). The thermal stability arises because the stratosphere contains O_3 (box 1.2), which warms the stratosphere by absorbing both IR radiation from below and UV radiation from above. Above the stratosphere is the *mesosphere*, which reaches to about 85 kilometers (53 miles). Here, temperature again decreases with altitude, reaching a minimum of about -100°C (-148°F) at the top of the mesosphere. Above that is the *thermosphere*, which constitutes only a minute fraction of the mass of the atmosphere but extends upward some 500 kilometers (300 miles).

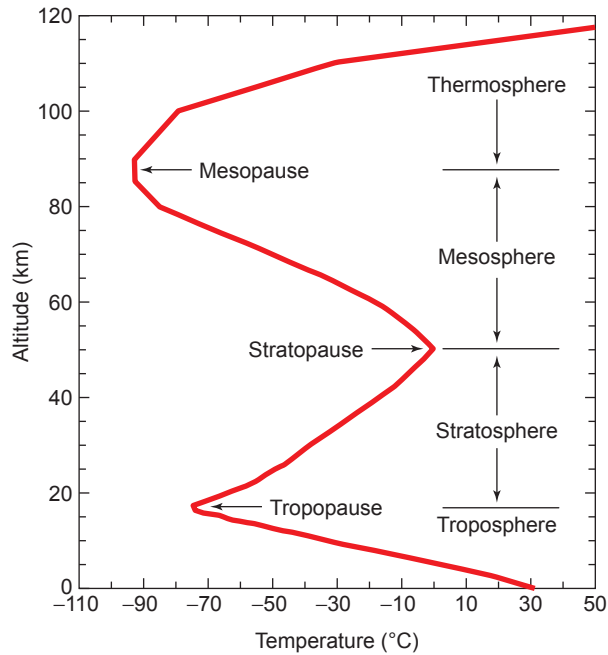


FIGURE 1.5 The layered structure of the atmosphere

The troposphere makes up about 80 percent of the total mass of the atmosphere, is where the weather occurs, and holds most of the planet-insulating greenhouse gases. The overlying stratosphere is relatively warm due to the presence of ozone, which absorbs ultraviolet radiation from above. With essentially no greenhouse gases, the mesosphere is the coldest part of the atmosphere. The vanishingly thin thermosphere is heated by solar radiation and by its interactions with the solar wind. (Redrawn from Hartmann, *Global Physical Climatology*)

Temperature increases with altitude in the thermosphere due to the atmosphere's absorption of solar radiation and the bombardment of the outer atmosphere by the solar wind, the continuous stream of protons and electrons given off by the Sun.⁸ During periods of strong solar activity, the temperature of air at the top of the thermosphere may reach 1,500°C (2,700°F), but this does not mean that it contains a lot of energy. In fact, the energy there is minuscule because there are so few gas molecules at this altitude, and a body would feel very cold.

Without incoming solar radiation, the temperature of the atmosphere would simply decrease smoothly with altitude as the density of air decreases. The layered temperature structure exists, however, because of how the atmosphere gains and loses heat. About half of the incoming solar radiation is absorbed by the ocean and land surfaces, which are, in turn, warmed by the absorbed energy. Depending on their temperatures, these warm surfaces emit IR radiation (chapter 5), the form of radiation we usually refer to as “heat.” Much of this heat is absorbed by the greenhouse gases in the troposphere, maintaining a surface temperature warmer than it would otherwise be (chapter 6). Additionally, *advective* (horizontal) and convective motions of the atmosphere transport latent and sensible heat away from the surface and stir them into the troposphere. The result is a surface temperature that averages about 15°C (59°F) over the entire globe, although obviously temperature varies substantially from one place to another. (A minimum temperature of -89.2°C [-128.5°F] was recorded at the Russian Vostok Station, Antarctica, in 1983, and a maximum temperature of 54.0°C

BOX 1.2

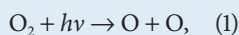
Stratospheric Ozone

Compared with the troposphere, dynamics in the stratosphere influence climate in secondary ways. We have already seen that the stratosphere is a fundamental part of the atmosphere, however, and stratospheric ozone is environmentally important. Let us therefore take a brief look at how stratospheric ozone is formed and destroyed.

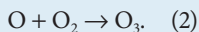
The ozone molecule consists of three atoms of oxygen (O_3).¹ The gas is not mysterious; it can be produced by an electric spark, so its distinctive, pungent odor is likely familiar to anyone who has played with electric trains or used an arc welder. The extremely low concentration of ozone in the atmosphere (typically 6 to 8 ppm in the stratosphere and much less elsewhere) disguises the fact that it either protects or harms life, depending on where it resides. About 90 percent of ozone in the atmosphere exists in a stratospheric layer between about 15 and 50 kilometers (9–30 miles) in altitude. Here, ozone is good because it absorbs most of the incoming solar UV radiation. If more of this radiation reached the surface, it would cause increased incidences of skin cancer, cataracts, and immune-system impairments, as well as damage crops and other plants. The term *ozone layer* is rather misleading because ozone is not really a “layer”; rather, the gas drifts and swirls around the stratosphere in changing concentrations, similar to the way clouds drift and swirl. The reason is that ozone is both rapidly formed and quickly destroyed in the stratosphere, and these processes operate at different rates in different regions. For example, an “ozone hole” (a region of low ozone concentration caused by human emissions of chlorofluorocarbons) develops in a seasonal cycle over Antarctica and intermittently over the Arctic in their respective late winters–early springs.

Ozone also exists in the lower troposphere as a component of the smog that hangs over many cities. This ozone is harmful because of its high reactivity, which, among other things, causes damage to living tissue. It is also a potent greenhouse gas. This distinction raises the obvious question of why the “bad” ozone near the surface does not rise to replace the “good” ozone in the stratosphere. One reason is that because ozone is so reactive, it does not remain in the troposphere long enough to mix. Another is that the troposphere contains only 10 percent of the total ozone, which is not enough to replenish the stratosphere.

This brings us to the central issue of how stratospheric ozone is created and destroyed.² Ultraviolet radiation is the key on both counts. In a process known as *photolysis*, UV photons dissociate the ordinary oxygen molecule (O_2) into two O atoms,



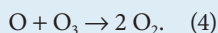
where $h\nu$ refers to the photon. Within a fraction of a second, the O atoms combine with O_2 to form O_3 ,



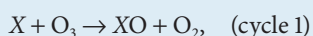
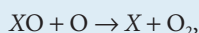
O_3 is then itself photolyzed to produce O_2 ,



While reactions 2 and 3 maintain a dynamic photolytic balance among O_2 , O_3 , and O , O_3 is also destroyed by O ,



In the nonpolar regions, certain natural compounds containing nitrogen (mainly nitric oxide [NO], originating from the breakdown of nitrous oxide [N_2O]), chlorine (mainly chloromethane [CH_3Cl], produced biologically in the oceans), and hydrogen (H or OH, produced by the breakdown of water vapor) are responsible for most of ozone destruction:

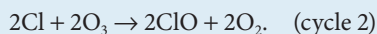
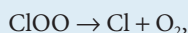
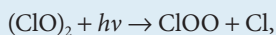
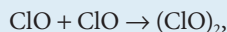


where X refers to any one of the natural compounds just mentioned. Note that X is not consumed in this cycle; rather, it serves as a catalyst. By summing the two reactions of cycle 1, it becomes apparent that the net result is simply equivalent to reaction 4.

Long before humans began polluting the atmosphere, the natural processes regulated the amount of ozone in the stratosphere, as ozone was created at about the same rate as it was destroyed. In the second half of the twentieth century, however, this natural balance started to change with the manufacture of certain ozone-destroying chlorine-, fluorine-, and bromine-containing compounds, known as *halocarbons*.³ Notable among these are the chlorofluorocarbons (CFCs), used primarily in air conditioners. They tend to be extremely stable and do not readily react with other chemicals in the lower atmosphere. In the stratosphere, however, UV radiation breaks them down to release free chlorine (Cl) atoms. The Cl then reacts with O_3 to form chlorine monoxide (ClO):



If each chlorine atom removed only one ozone molecule, CFCs would hardly threaten the ozone layer. However, reaction 5 initiates another catalytic cycle:



Summing these reactions shows that in net, cycle 2 destroys two O_3 molecules and produces three O_2 molecules. The process is also catalytic: ClO persists and is thus available to

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(Continued)

decompose other ozone molecules. In this way, a single CFC molecule can destroy thousands of ozone molecules. An analogous cycle exists for bromine.

Cycle 2 is implicated in the formation of the ozone hole. In the Antarctic, winter stratospheric temperatures drop below -80°C (-112°F), and water vapor sublimates to form microscopic ice particles that make up *polar stratospheric clouds*. The particles provide surfaces on which chlorine molecules (Cl_2) catalytically form. Unlike Cl , Cl_2 is inert—it does not react with anything. Thus when stratospheric clouds form over Antarctica in the winter, Cl_2 builds up, but ozone is not affected. When the spring sunlight strikes the Antarctic stratosphere in late August, however, UV radiation photolyzes Cl_2 to form Cl , restarting the destructive ozone cycle 2 and creating the “ozone hole.” The ozone hole now regularly appears above Antarctica each year beginning in the austral spring (figure B.1.2.1).⁴

Although the use of halocarbons has been curtailed by international treaty, another complication has significant bearing on how fast the ozone layer will recover. Chlorine can exist in the stratosphere as the chlorinated nitrite ClONO_2 . This compound is, fortunately, inert, but ice particles incorporate nitrogen to make a compound known as nitric acid trihydrate (NAT). The NAT-bearing ice particles enlarge in stratospheric clouds and eventually settle

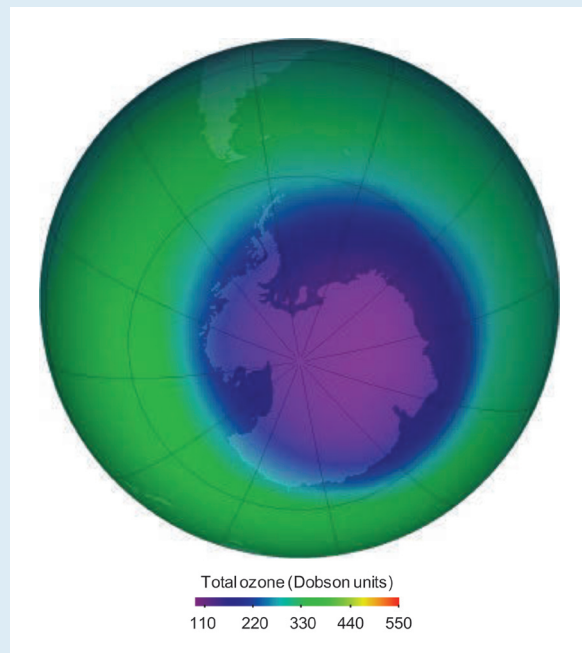


FIGURE B1.2.1

The hole in the ozone layer (*purple*) over the Antarctic as it appeared in October 2006. Ozone concentration is given in Dobson Units. One Dobson Unit is defined as a layer of pure ozone 0.01 millimeter (3.9×10^{-4} inches) in thickness at standard temperature and pressure. Thus 300 Dobson Units is a 3-millimeter (0.1-inch) thick layer of pure ozone distributed through the entire atmosphere column. (NASA/Goddard Space Flight Center, “NASA Ozone Watch,” <http://ozonewatch.gsfc.nasa.gov/monthly/SH.html>)

out of the stratosphere. By this mechanism, the stratosphere becomes “denitrified.” The loss of nitrogen means that none is available to form ClONO_2 , resulting in relatively more chlorine available to destroy ozone.⁵

Because stratospheric clouds form every winter over Antarctica, the region is always effectively denitrified. But in the Arctic, the winters are warmer, and until recently stratospheric clouds formed only rarely. Now, with the increased amount of greenhouse gases in the troposphere, the Arctic stratosphere is cooler. The lower temperatures mean that clouds of NAT-bearing ice particles form more frequently, causing denitrification of the Arctic stratosphere, which, in turn, causes an ozone hole to open up in the Arctic winter–spring where none existed before the advent of CFCs.⁶

Fortunately, chlorine does not remain in the stratosphere forever. It eventually reacts with compounds such as CH_4 to produce stable hydrochloric acid (HCl), which is also incorporated into ice particles and settles into the troposphere to be washed out by rain. As the use of halocarbons is being phased out, their abundance in the stratosphere has begun to decline from a maximum in the early 1990s, and the ozone layer will eventually repair itself if the world’s governments abide by their treaties. Current projections suggest that pre-1980 levels of ozone will return by the latter part of the twenty-first century.⁷ Needless to say, substantial issues remain. For example, the environmental impacts of the compounds that are being developed to replace CFCs are not completely known, and some sources of chlorine have yet to be regulated. In the meantime, the large decline in stratospheric ozone abundance has had other effects. In particular, it, along with the rising contents of greenhouse gases in the troposphere, has caused a steady cooling of the stratosphere.

1. The word *ozone* is a compound of the Greek word *ózo* (a smell).

2. This account is largely taken from J. Pyle et al., “Ozone and Climate: A Review of Interconnections,” in *Special Report on Safeguarding the Ozone Layer and the Global Climate System: Issues Related to Hydrofluorocarbons and Perfluorocarbons*, ed. B. Metz et al. (Cambridge: Cambridge University Press, 2005), 83–132.

3. The damaging effect of CFCs on ozone was discovered by M. J. Molina and F. S. Rowland, “Stratospheric Sink for Chlorofluoromethanes: Chlorine Atom Catalysed Destruction of Ozone,” *Nature* 249 (1974): 810–812; and P. J. Crutzen, “Estimates of Possible Future Ozone Reductions from Continued Use of Fluoro-Chloromethanes (CF_2Cl_2 , CFCl_3),” *Geophysical Research Letters* 1 (1974): 205–208. For their discovery, the three shared the Nobel Prize in Chemistry in 1995.

4. See, for example, J. C. Farman, B. G. Gardiner, and J. D. Shanklin, “Large Loss of Total Ozone in Antarctica Reveal Seasonal ClO_x/NO_x Interaction,” *Nature* 315 (1985): 207–210; and S. Solomon, “Stratospheric Ozone Depletion: A Review of Concepts and History,” *Reviews of Geophysics* 37 (1999): 275–316.

5. For example, D. W. Fahey et al., “The Detection of Large HNO_3 -Containing Particles in the Winter Arctic Stratosphere,” *Science* 291 (2001): 1026–1031.

6. G. L. Manney et al., “Unprecedented Arctic Ozone Loss in 2011,” *Nature* 478 (2011): 469–475.

7. Pyle et al., “Ozone and Climate.”

[129.2°F] was recorded in Death Valley, California, in 1913 and in Mitribah, Kuwait, in 2016.) At the same time, the troposphere cools from the top as it radiates heat upward.

From these processes, together with convective mixing and adiabatic temperature changes, the troposphere reaches an overall “radiative–convective equilibrium” whereby temperature decreases with increasing altitude at a global mean rate of 6.5°C (11.7°F) per kilometer.⁹ The change in temperature with altitude in the troposphere is known as the *lapse rate*

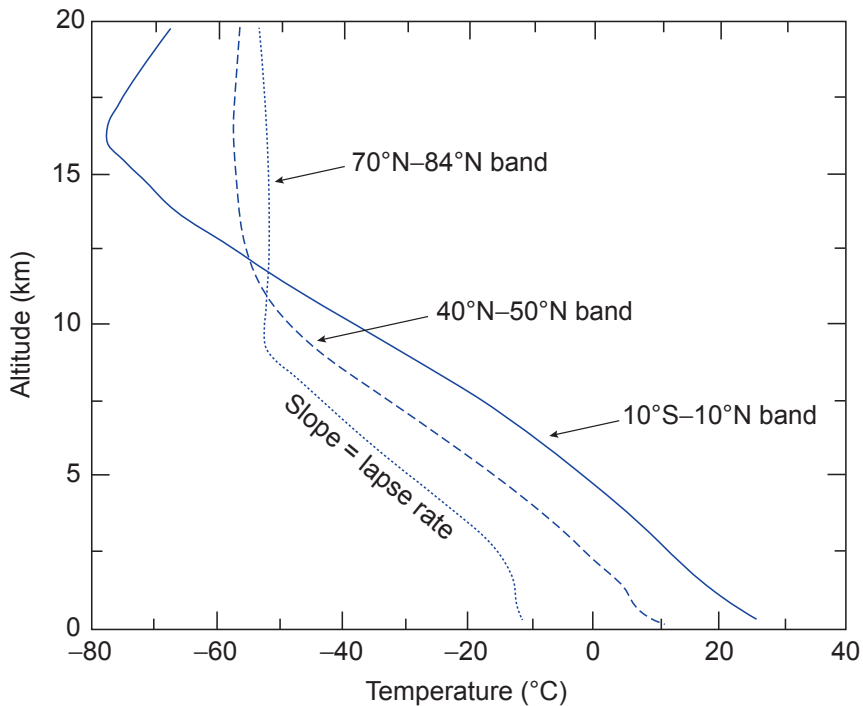


FIGURE 1.6 Profiles of mean annual temperature as a function of altitude (lapse rates) at different latitudes

As a result of the way the troposphere gains and loses heat and the way that heat is convectively distributed within it, the troposphere tends to an overall “radiative–convective equilibrium,” whereby temperature decreases with altitude. The changes in the mean annual lapse rate and in the height and thickness of the tropopause with latitude mainly reflect the fact that incoming radiation is greater than outgoing radiation in the tropics, while the opposite is true at the high latitudes. (Redrawn from Hartmann, *Global Physical Climatology*)

(figure 1.6). Its manifestations are the common experience of cooler conditions at higher elevations and snow-capped mountain peaks.

CONSEQUENCES OF A STABLE ATMOSPHERIC STRUCTURE

The structure of the atmosphere, like its composition, is essential for life, basically because it keeps the planet from drying out. The very cold tropopause and the even colder mesopause (the top of the mesosphere) trap water vapor below because the cold air can hold very little water. Any water vapor that rises to these heights sublimates (changes state from vapor directly to ice) to form ice particles that eventually settle out, keeping the upper atmosphere dry. Solar radiation dissociates water vapor that does reach the upper atmosphere, creating free hydrogen (H_2). The H_2 molecule has so little mass and at high altitudes sufficient kinetic energy that it is able to escape Earth’s gravitational field to be lost to space. Had this happened to an appreciable degree for an appreciable time, it is conceivable that Earth could have become completely dehydrated, in which case the surface would now be devoid of both water and life. Indeed, this process may have taken place on our neighboring planet Mars.

The structure and stability of the atmosphere also have consequences for weather and pollution. Weather originates mostly from thermally driven motions in the troposphere. Although the air of the troposphere may mix into the base of the stratosphere, the mixing of stratospheric air into the rest of the atmosphere is slow. For this reason, when pollutants do reach the stratosphere, they tend to stay there and are quickly transported around the globe. As we shall see (chapter 6), this outcome happens in a spectacular way when volcanic eruptions inject sulfur into the stratosphere.

The General (Global) Circulation of the Atmosphere

Mariners are familiar with the ocean-wide wind systems that have been pushing ships back and forth across the oceans for centuries. The easterlies (winds are commonly described based on the direction from which they blow), also known as the *trade winds*, carried them westward. These winds are driven by the difference in atmospheric pressure between the so-called *subtropical highs* at about the 30° latitudes and a band of low pressure at the equator. In the early days of sailing, the return trip was made possible by the westerlies. These winds blow between about the 30° and 60° latitudes and are driven by the difference in atmospheric pressure between the subtropical highs and the bands of low pressure known as the *subpolar lows*. As suggested by the easterlies and westerlies, the winds are strongest parallel to latitude, not in the north–south (longitudinal, or “meridional”) direction. The directionality of the winds has to do with Earth’s rotation and is the result of the Coriolis effect. This effect interacts with the north–south circulation patterns transporting warm air to high latitudes to give us the observed global wind patterns.

THE CORIOLIS EFFECT

The *Coriolis effect* is the result of Earth’s rotation and is named for the nineteenth-century French mathematician Gaspard-Gustave de Coriolis (1792–1843), who first explained how Earth’s rotation influences the movement of fluid. In particular, the Coriolis effect is the apparent deflection of a moving object, such as air or water masses, from the perspective of a fixed point on the rotating Earth. The deflection is to the right in the Northern Hemisphere and to the left in the Southern Hemisphere when facing the direction of movement.

To understand the Coriolis effect, let us start by imagining two children on opposite sides of a stationary circular platform tossing a ball back and forth (figure 1.7). As long as the children toss the ball directly to each other, the ball passes in a straight line over the center of the platform to arrive at its intended target. Next, imagine that the platform is rotating counterclockwise. Now when one child tosses the ball directly toward the child on the other side of the platform, from the perspective of the children on the platform, the ball deflects to the right of its intended target. The reason is that the ball is unaffected by the rotation of the platform because it is in the air and thus travels in a straight line, while simultaneously the children rotate counterclockwise because they are sitting on the platform.

The rotation of Earth is counterclockwise when observed from space looking down on the North Pole, so an object moving from the North Pole toward the equator is deflected to the

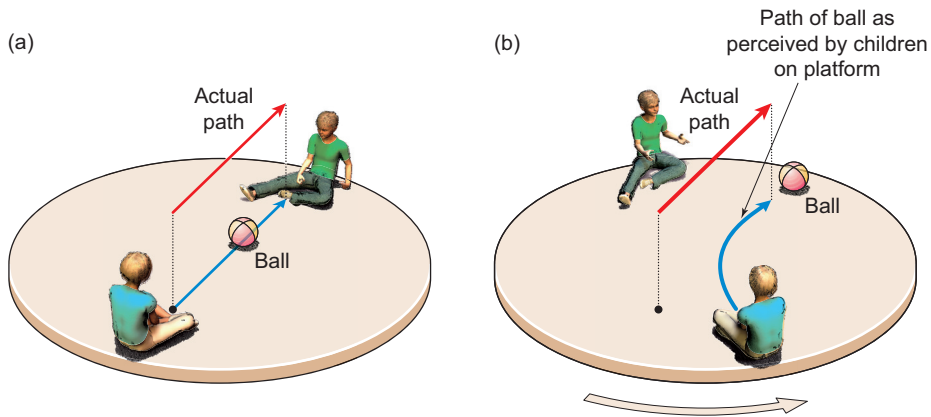


FIGURE 1.7 The Coriolis effect on a flat, rotating platform

Two children on opposite sides of a platform toss a ball back and forth. When the platform is stationary (a), the ball follows a straight line between the children, passing over the center of the platform. When the platform is rotating counterclockwise (b), the ball appears to deflect to the right relative to its intended path and misses the child who is to catch it. This is because the ball travels through the air in a straight line and is unaffected by the rotation of the platform, but the children, who are sitting on the platform, rotate counterclockwise.

right. In contrast, the rotation at the South Pole is clockwise, and the deflection of an object moving toward the equator is to the left. More generally, the Coriolis force causes objects to deflect to the right relative to the direction of their paths in the Northern Hemisphere and to the left relative to the direction of their paths in the Southern Hemisphere (figure 1.8). Finally, the Coriolis force is important only to phenomena involving high velocities over large spatial scales of hundreds of kilometers. (Contrary to popular belief, it does not determine the way that water drains in your shower or toilet!)

The Coriolis force also causes winds to blow counterclockwise around low-pressure systems in the Northern Hemisphere and clockwise around such systems in the Southern Hemisphere, directionalities termed *cyclonic* flow. The opposite happens around high-pressure systems: winds around them flow clockwise in the Northern Hemisphere and counterclockwise in the Southern Hemisphere and are said to be *anticyclonic*. As we shall see, the cyclonic and anticyclonic flows are particularly important in the mid-latitudes.

HADLEY CELLS AND THE INTERTROPICAL CONVERGENCE ZONE

The English mathematician George Hadley (1685–1768) was the first to explain the persistent trade winds. In 1735, he pointed out that more solar energy reaches the surface at the equator than at the poles because Earth is spherical. More precisely, in the tropics, the amount of incoming solar radiation is greater than the amount of outgoing planetary radiation, while the opposite is true near the poles. The plethora of heat in the tropics causes warm equatorial air to rise to the top of the troposphere and flow outward toward the poles, to be replaced by cool air flowing toward the equator from the poles. This circulation transfers a massive amount of heat from the equator to the poles and is the fundamental process by which Earth achieves a balance between its incoming and its outgoing energy.

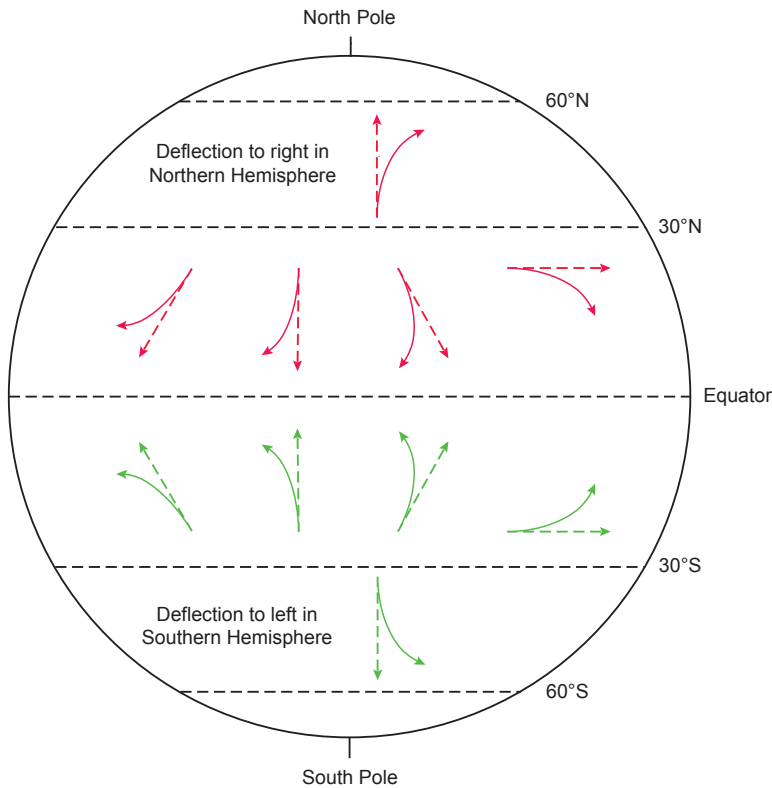


FIGURE 1.8 The deflection of winds in the Northern and Southern Hemispheres due to Coriolis forces

In the Northern Hemisphere, the deflection is always to the right, while in the Southern Hemisphere, it is always to the left when facing the direction of flow.

The transfer begins with a convective circulation in each hemisphere in what are known as the *Hadley cells*. Warm, moist air convectively rises near the equator to form the upward limb of the Hadley cells (figure 1.9). Near the top of the troposphere, the air flows poleward, while being deflected eastward by the Coriolis force. At about the 20° to 30° latitudes, the now relatively cold and dry air descends to create zones of high pressure—the subtropical highs. Because the air is sinking, the subtropical highs correspond to dry regions of low rainfall and many of the world’s great deserts (figure 1.10). The return flow (which is deflected by the Coriolis force) constitutes the easterly trade winds.

The trade winds converge at the *Intertropical Convergence Zone* (ITCZ), the equatorial zone of low pressure, convection, rainfall, and consequent transfer of substantial latent heat to the high troposphere between the Northern and Southern Hemisphere Hadley cells (figure 1.10). The transfer of heat and moisture from the surface to the upper troposphere occurs chiefly through cumulonimbus clouds, the towering clouds of thunderstorm systems. Over the oceans, the ITCZ is marked by zones of light and variable winds known as the doldrums.

In reality, Hadley circulation is far more complex than is suggested by this idealized construct. First, the Hadley cell in the winter hemisphere is vastly dominant over the cell in the summer hemisphere, with the ITCZ being displaced into the latter (figure 1.11). Thus during

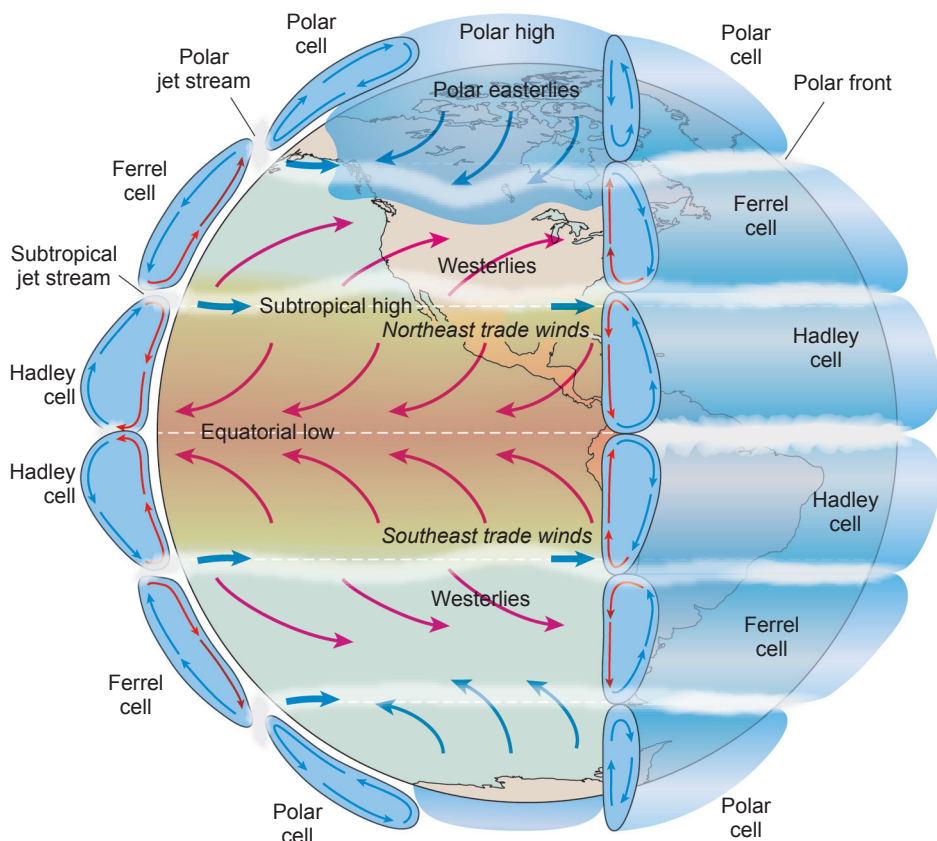


FIGURE 1.9 Idealized surface and global wind pattern

Warm air rises near the equator to create a zone of low pressure and the upward limbs of the Hadley cells. Downward flow of the Hadley cells results in two zones of high pressure at about the 20° to 30° latitudes, where most of the world's great deserts are located; the return flow constitutes the trade winds. The Coriolis effect breaks up simple Hadley cell circulation to create the Ferrel cells, which are dominated by eddy circulations (*not shown*) and result in westerly winds. The Polar cells are caused by the descent of dry air near the poles. (Redrawn from F. K. Lutgens and E. J. Tarbuck, *The Atmosphere: An Introduction to Meteorology*, 8th ed. [Upper Saddle River, N.J.: Prentice Hall, 2000])

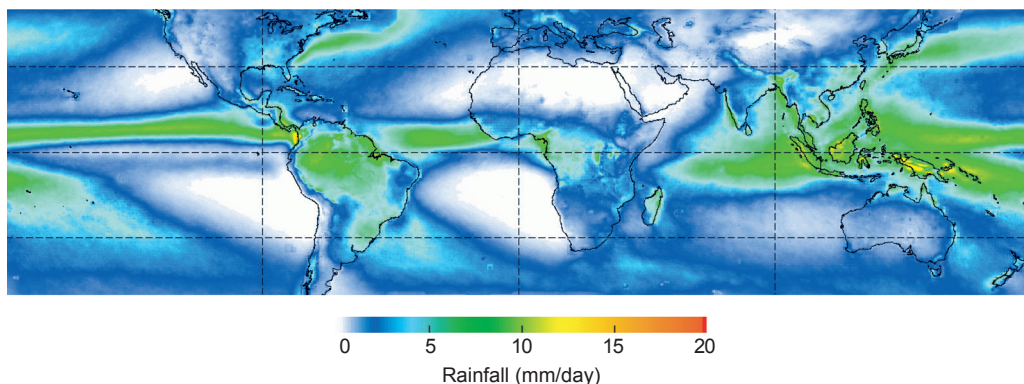


FIGURE 1.10 The global mean daily rainfall based on satellite observations, 1998–2011

The equatorial belt of high rainfall marks the position of the Intertropical Convergence Zone, and the subtropical belts of low rainfall correspond to the subtropical highs and corresponding desert regions. (NASA/Goddard Space Flight Center, "Climate Change: Trends and Patterns," <http://pmm.nasa.gov/science/climate-change>)

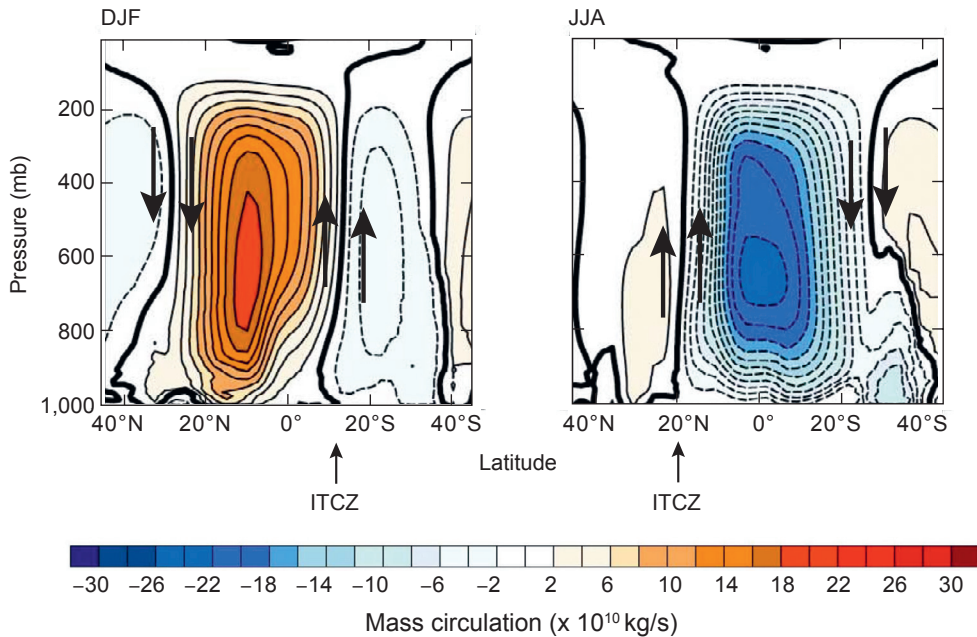


FIGURE 1.11 Meridional cross sections showing the change in Hadley circulation from winter to summer, 1979–2008

The data are the average of the annual zonal mean mass circulation from winter (DJF) to summer (JJA), based primarily on satellite data. Positive values (*red shades*) represent counterclockwise flow; negative values (*blue shades*) represent clockwise flow. The heavy black arrows indicate the direction of vertical flow. The seasonal changes in the position of the ITCZ are also shown. (Redrawn from J. P. Stachnik and C. Schumacher, “A Comparison of the Hadley Circulation in Modern Reanalyses,” *Journal of Geophysical Research* 116 [2011], doi: 10.1029/2011JD016677)

the Northern Hemisphere winter, the ITCZ is typically at about 5°S latitude, and the descending air is at about 20°N, while during the winter in the Southern Hemisphere, the ITCZ is at about 10°N. In detail, however, the ITCZ undulates through the tropics with longitude and with the seasons (figure 1.12). In other words, the Northern and Southern Hemisphere Hadley cells oscillate in strength and size through the seasons, and the ITCZ and subtropical high-pressure zones likewise undulate back and forth. This behavior reflects the fact that the “thermal equator,” or the latitudinal zone where Earth is heated most, shifts with the seasons.

The tropics are also characterized by a set of large-scale, east–west (that is, along the equator) convection cells known as Walker circulation. The most prominent of these is over the Pacific Ocean. The easterly flow of air across the equatorial Pacific convectively rises in the western equatorial Pacific, and the return flow near the top of the troposphere sinks to deliver dry, relatively cool air to the central and eastern equatorial Pacific. Similarly, albeit less powerful, east–west circulations exist in the equatorial Atlantic Ocean and elsewhere along the equator. Walker circulation is part of the phenomenon known as the El Niño–Southern Oscillation (ENSO), an important coupling of ocean and atmospheric circulation. Accordingly, we shall return to the Walker circulation and explore ENSO in more detail in the context of interactions between the ocean and the atmosphere (chapter 3).

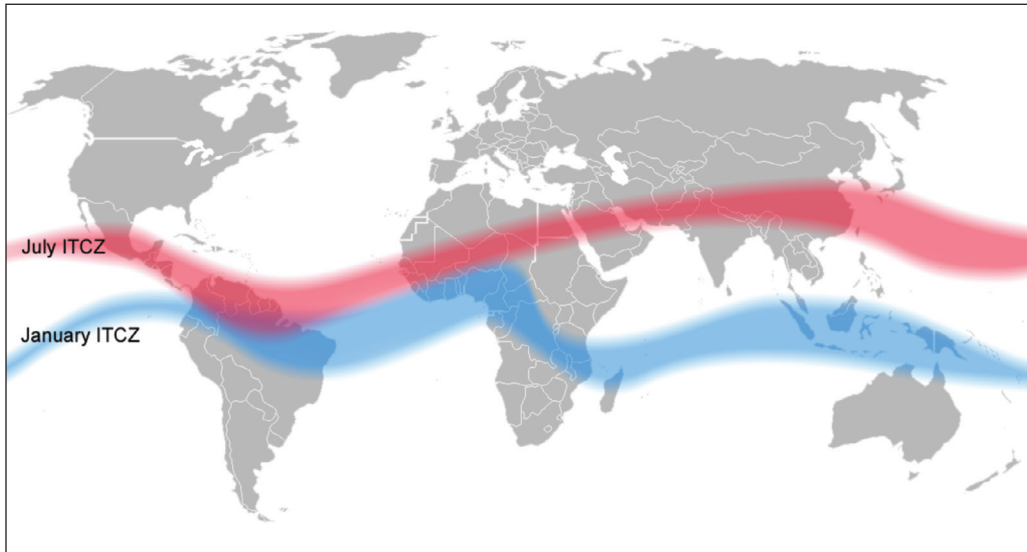


FIGURE 1.12 *The seasonal change in the location of the ITCZ on the globe*

The ITCZ shifts in latitude with the seasons because it represents Earth's "thermal equator." (M. Halldin, in "Intertropical Convergence Zone," Wikipedia, https://en.wikipedia.org/wiki/Intertropical_Convergence_Zone)

POLAR AND FERREL CELLS

To understand what happens poleward of the Hadley cells, it is useful to distinguish mid- to upper-tropospheric and surface patterns of circulation. Poleward of the subtropical highs, the mid- to upper-tropospheric winds are dominantly westerly in both hemispheres. In the Southern Hemisphere, they form a year-round, almost hemisphere-wide cyclonic vortex centered over Antarctica, as reflected by variations in atmospheric pressure (figure 1.13a). Near the surface, air descends near the South Pole to form a weak surface high-pressure region surrounded by a belt of low pressure at about 60°S, where the air that spreads out from the high pressure over Antarctica rises, resembling the idealized *Polar cell* depicted in figure 1.9.

In the Northern Hemisphere, a cyclonic vortex of mid- to upper-tropospheric westerlies similar to that in the Southern Hemisphere also exists, especially in the winter, but the Northern Hemisphere vortex is more deformed by mountain ranges and surface thermal discontinuities (see figure 1.13b). For example, the mid-tropospheric pressure surface displays prominent valleys at 150°E and 70°W of longitude. The surface wind patterns are also much more complex. Instead of a continuous belt of low pressure around the pole, winter lows appear near the Aleutian Islands and Iceland, and shallow highs appear over Siberia and western Canada.

Between the Hadley and Polar cells in each hemisphere is a zone of surface (and upper-tropospheric) westerlies known as the *Ferrel cells* that result from the breakdown of Hadley circulation by Coriolis forces. They are commonly described as having a mean meridional air flow opposite that of the Hadley and Polar cells, with cool air rising at about the 50° latitudes and warm air sinking at the 30° to 40° latitudes (see figure 1.9). As such, they are "thermodynamically indirect" because they transport energy from cold to warm regions. This meridional flow is weak, however, and nearly all transport of heat and moisture poleward

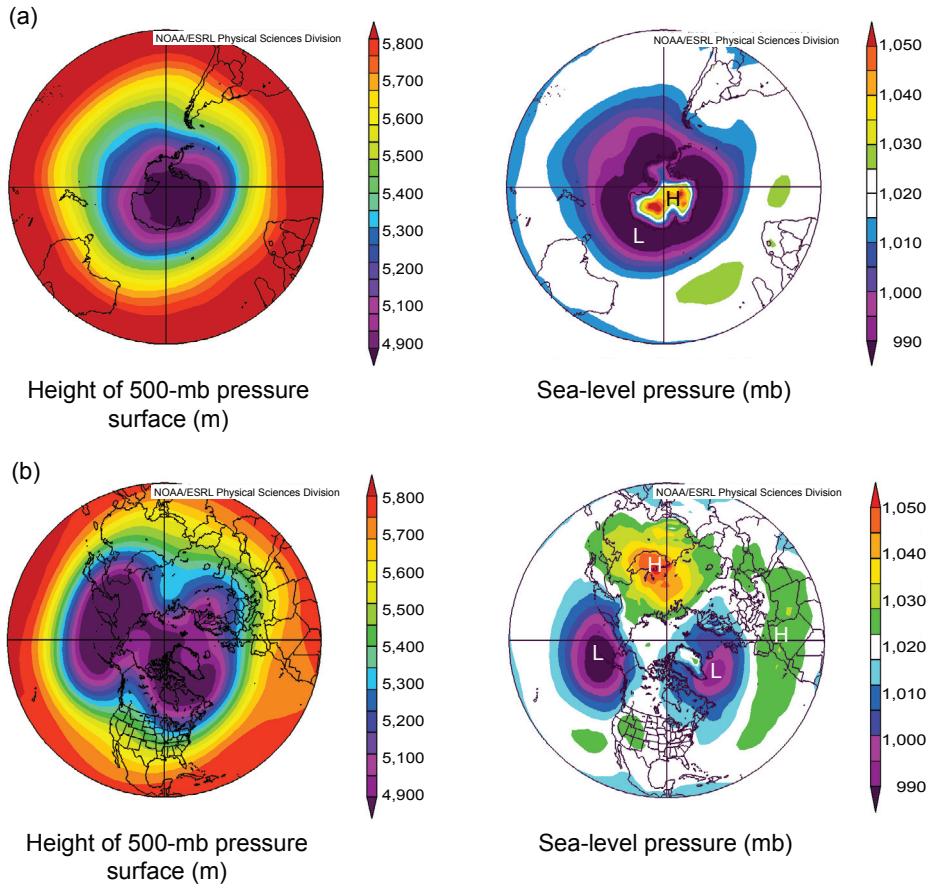


FIGURE 1.13 Heights of the 500-millibar pressure surfaces and sea-level pressures for the Southern Hemisphere and Northern Hemisphere winters

(a) Mean height above sea level of the 500-millibar (mb) atmospheric pressure surface (left) and mean sea-level pressure (right) for the Southern Hemisphere for July 2016. The 500-mb surface indicates that in the Southern Hemisphere winter, the mid- to upper-tropospheric winds generally form a single, enormous cyclonic vortex centered on Antarctica. At the surface, the Polar cell is typically defined by a weak high (H) over Antarctica surrounded by a belt of low pressure (L). (b) Mean height above sea level of the 500-mb atmospheric pressure surface (left) and mean sea-level pressure (right) for the Northern Hemisphere for January 2016. The landmasses deform the high-altitude cyclonic flow, and the mean surface pressure displays lows near the Aleutians and Iceland and shallow highs over Siberia and western Canada. The 500-mb surface corresponds to a height above sea level of about 5.6 kilometers (3.5 miles). (Redrawn from NOAA, Earth System Research Laboratory, Physical Sciences Division, “6-Hourly NCEP/NCAR Reanalysis Data Composites,” <https://www.esrl.noaa.gov/psd/data/composites/hour/>)

and of cool air equatorward is instead accomplished by *eddy circulation* (figure 1.14). Eddies are a sequence of transient cyclone/anticyclones, or weather systems, that march across the mid-latitudes from west to east. In the cyclonic flow around low-pressure centers, warm and moist air rises, and this pulls in and entrains warmer surface air from the lower latitudes. The Ferrel cells were named for the American meteorologist William Ferrel (1817–1891), who recognized the important role of mid-latitude eddies in atmospheric heat transport.

At about the 60° latitudes, the westerlies encounter the polar air masses along the so-called *polar front*, a zone of unstable and sometimes severe weather that arises from the colliding

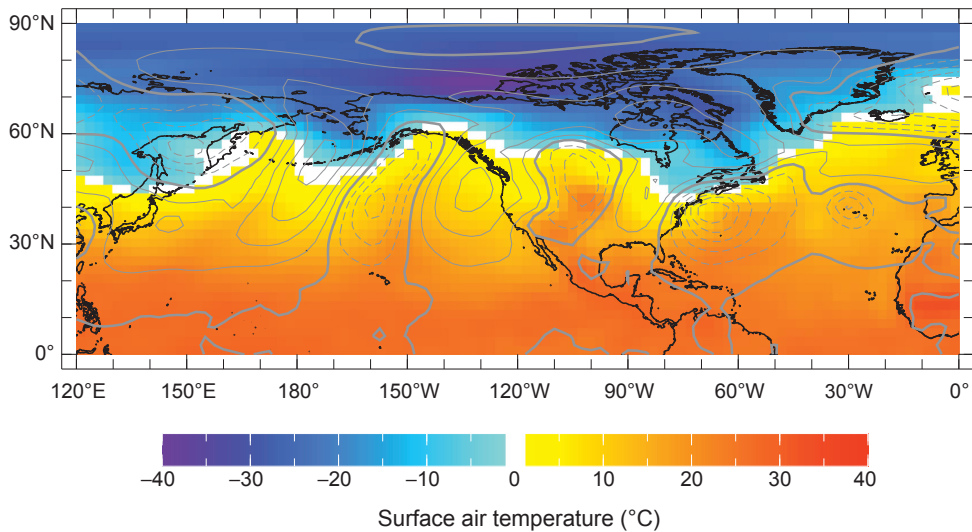


FIGURE 1.14 Mid-latitude eddies represented by contours of daily mean sea-level pressure anomaly [relative to the 1979–2014 mean] and by surface temperature

For the pressure field, the contour interval is 5 mb; the thick gray lines represent zero anomaly; and the dashed and thin solid contours represent positive and negative anomalies, respectively. The data illustrate the counter-clockwise flow and mixing of air around low-pressure eddies, indicated by valleys in the sea-level pressure field, that draw warm air from the subtropics into the mid-latitudes. The daily data are for April 1, 1997. (NOAA NCEP-DOE Reanalysis-2 dataset; M. Kanamitsu et al., “NCEP-DOE AMIP-II Reanalysis [R-2],” *Bulletin of the American Meteorological Society* 83 [2002]: 1631–1643)

air masses. At the polar front, the mid- to upper troposphere is characterized by a quasi-permanent to slow-moving meridional meander in westerly air flow. The meandering air flow gives rise to what are known as *planetary waves*, or *Rossby waves*, after the noted meteorologist Carl-Gustaf Rossby (1898–1957) (Historical Note). In the winter, there are typically three or four waves around the hemisphere with wavelengths of several thousand kilometers. Rossby waves expand and contract in amplitude and may form smaller, transient eddies embedded in the generally upper-tropospheric westerly flow field. The larger of these tend to steer the weather patterns and thus account for much of the variability in mid-latitude weather. The transient eddies evolve to organize themselves into cyclone/anticyclone cells and decay as new waves take their place. In the winter, the stationary and transient eddies are responsible for nearly all of the meridional transport of heat and moisture. The energy flux associated with transient eddies gives rise to mid-latitude storm systems that also progress from west to east.

JET STREAMS

This brings us to the *jet streams*. Jet streams are high-speed, horizontal rivers of wind some hundreds of kilometers wide and several kilometers thick (figures 1.15 and 1.16). They form at the top of the troposphere, where the meridional temperature gradients are largest. Thus the polar jet streams, with peak winds of about 300 kilometers (190 miles) per hour, oscillate back and forth with Rossby waves, marking the polar front in the upper troposphere. These



HISTORICAL NOTE

Carl-Gustaf Arvid Rossby and Meteorology

Carl-Gustaf Rossby (1898–1957) was one of a group of notable Scandinavian researchers who worked with the Norwegian meteorologist Vilhelm Bjerknes (1862–1951) at the University of Bergen in the early twentieth century and would go on to shape modern meteorology and oceanography.¹ The main approach of the so-called Bergen school was to find theoretical ways of describing the motions of the atmosphere based on the laws of hydrodynamics and thermodynamics and apply them to weather forecasting.²

Growing up in Stockholm, Rossby received a traditional education. He earned a degree in mathematical physics at the University of Stockholm in 1918, but after hearing a lecture by Bjerknes, and apparently bored with Stockholm, he moved to the newly established Geophysical



Carl-Gustaf Rossby. (University of Chicago Photographic Archive [apf1-10912], Special Collections Research Center, University of Chicago Library)

Institute in Bergen. Realizing that he needed more training in mathematics, he returned to Stockholm in 1921 to take a position as a meteorologist at the Swedish Meteorological and Hydrological Institute, and to continue his studies, earning a Filosofie Licentiat (at the time, equivalent to a doctoral degree in the United States) in mathematical physics at the University of Stockholm in 1925. He also gained valuable practical experience on summer cruises. In 1923, for example, he sailed on a Norwegian ship to East Greenland to establish a weather station for the Bergen group. The trip turned harrowing, however, when the ship became trapped in the ice pack for two months and almost sank.

In 1925, Rossby, apparently unfulfilled by routine weather forecasting, received a fellowship from the Sweden-America Foundation to go to the United States, where he joined the United States Weather Bureau. Based in part on his practical experience in weather forecasting, Rossby had become a proponent of the “polar front theory,” which explains the cyclonic circulation that develops at the boundary between warm and cold air masses. Rossby brought with him his enthusiasm and desire to apply the theory, but, according to one biographer, “the Central Office of the Weather Bureau was not a pleasant place for the twenty-seven-year-old Swede. He encountered unexpected hostility to the polar-front theory and to air-mass analysis and other approaches derived from it . . . [and] there was no one to talk to about hydrodynamics, atmospheric flow, and mathematical physics.”³

In a case of being in the right place at the right time, Rossby accepted a position at the Daniel Guggenheim Fund for the Promotion of Aeronautics, an organization devoted to

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supporting aviation.⁴ There, he developed an “airway system” (a set of weather-observation points along the route) for flights between Los Angeles and San Francisco. In 1928, the Guggenheim Fund established the Department of Aeronautical Engineering at the Massachusetts Institute of Technology, and Rossby became an associate professor at MIT. His job was to develop a curriculum in meteorology, which soon evolved into the first Department of Meteorology in the country.

Rossby remained at MIT for 11 years, during which time he married Harriot Alexander (1929), with whom he had three children, and became an American citizen (1939). He developed his most important theoretical contributions describing the motion of long waves in the upper troposphere (known as Rossby waves), became recognized as perhaps the world’s leading meteorologist, and started several new journals. Just as important, “those who studied under him practically worshiped him. They were participating in his great crusade—to bring modern meteorology to the United States.”⁵

In 1939, Rossby returned to the Weather Bureau as the assistant chief for research, presumably to place weather forecasting on a sounder scientific foundation. After two years, he left to head the new Department of Meteorology at the University of Chicago. During the war years, he organized the training of literally thousands of meteorologists for the military and traveled constantly to different theaters of the war to manage and advise on weather-forecasting efforts—for example, by helping to establish an institute of tropical meteorology. Rossby also served as president of the American Meteorological Society for two years, transforming it into a professional organization.

In 1947, Rossby accepted the chair of the Institute of Meteorology, which had been set up for him at the University of Stockholm, where he remained until his death ten years later. There he started yet another journal (*Tellus*), set up the first European center devoted to forecasting using numerical computer models, and began thinking about atmospheric chemistry. This thinking must have given him foresight into the environmental movement that was soon to come: in 1956, he appeared on the cover of the December 17 issue of *Time*, where in the accompanying article he admonished, “We should have a great deal of respect for the planet on which we live.”⁶

1. Much has been written about Rossby, including a series of papers published in the *Bulletin of the American Meteorological Society* in the late 1990s. This historical note is based primarily on H. B. Byers, *Carl-Gustaf Arvid Rossby, 1898–1957: A Biographical Memoir* (Washington, D.C.: National Academy of Sciences, 1960); and N. A. Phillips, “Carl-Gustaf Rossby: His Times, Personality, and Actions,” *Bulletin of the American Meteorological Society* 79 (1998): 1097–1112.

2. R. Jewell, “The Bergen School of Meteorology,” *Bulletin of the American Meteorological Society* 62 (1981): 824–830.

3. Byers, *Carl-Gustaf Arvid Rossby*, 252–253.

4. Daniel Guggenheim Fund for the Promotion of Aeronautics, Manuscript Division, Library of Congress, http://findingaids.loc.gov/db/search/xq/searchMfero2.xq?_id=loc.mss.eadmss.ms011196&_faSection=overview&_faSubsection=did&_dmdid=. The fund was active from 1926 to 1930. It dispersed about \$3 million, mostly to various universities to establish schools of aeronautics.

5. Byers, *Carl-Gustaf Arvid Rossby*, 257.

6. As reported in Phillips, “Carl-Gustaf Rossby,” 1107.

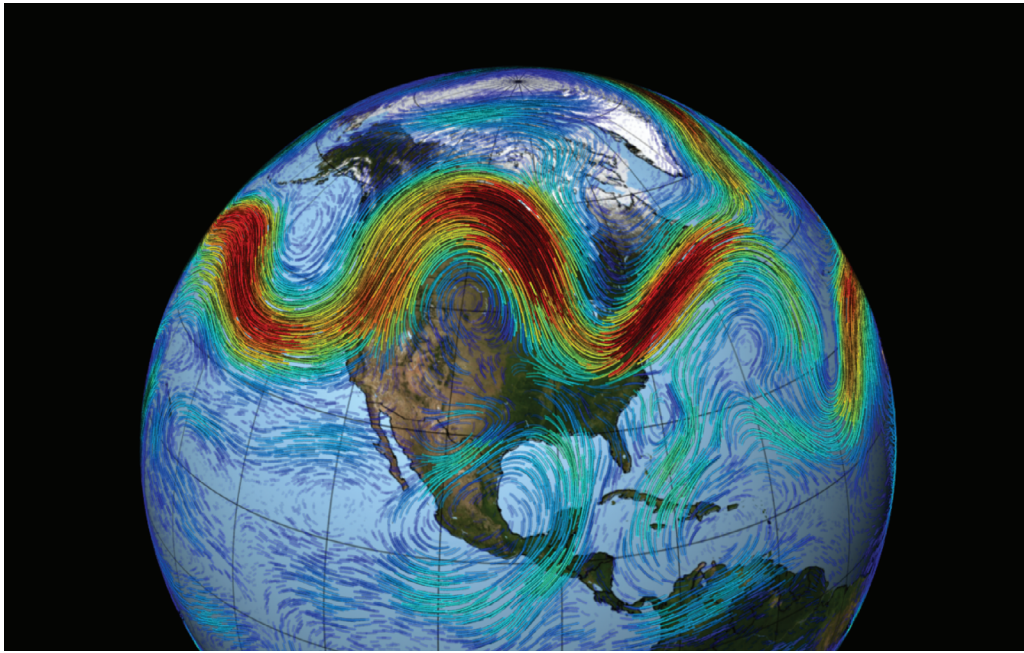


FIGURE 1.15 Visualization of oscillations in the polar jet stream

The undulating paths of the polar jet stream influence weather, especially in the winter, between about the 45° and 60° latitudes. Wind velocity grades from fastest (*red*) to slowest (*blue*). (K. Sharghi, “Aerial Superhighway,” February 7, 2012, NASA/Goddard Space Flight Center, <https://svs.gsfc.nasa.gov/10902>, which includes an animated version of this image)

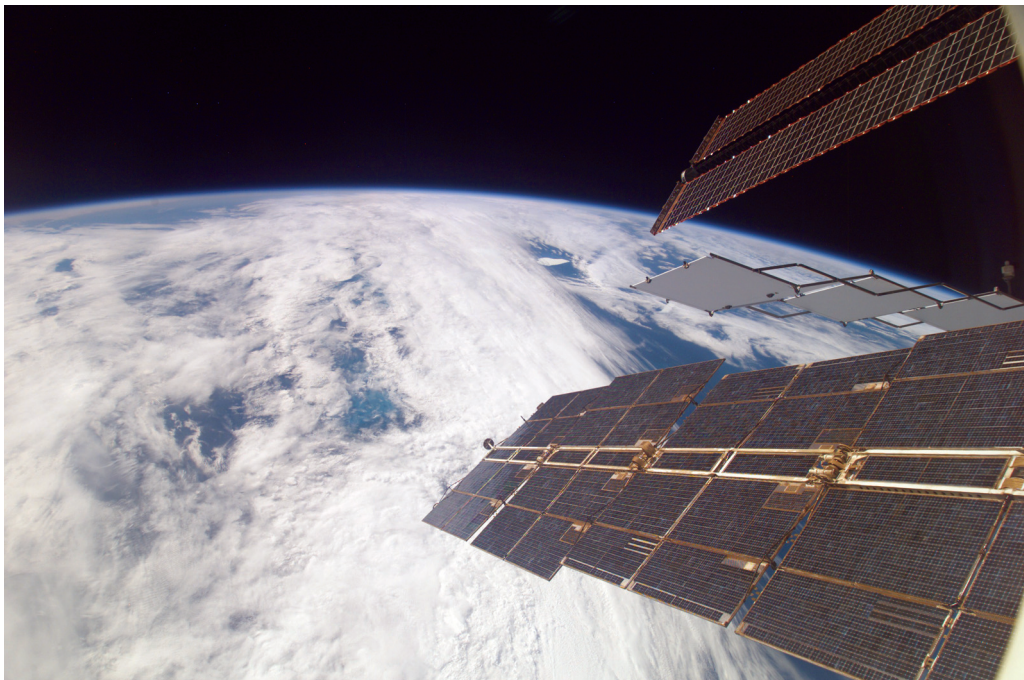


FIGURE 1.16 The Northern Hemisphere polar jet stream

A linear belt of clouds defines the polar jet stream over Newfoundland, Canada, as observed from the International Space Station. (NASA, Johnson Space Center, <https://eol.jsc.nasa.gov/SearchPhotos/photo.pl?mission=ISS008&roll=E&frame=12443>)

slow undulations affect weather patterns between about the 45° and 60° latitudes and are especially strong during the winter. Another set are the subtropical jet streams at the 30° latitudes. These locations correspond to the downward limbs of the Hadley cells, which account for the sudden drops in the tropopause mentioned earlier. Wind speeds up to 540 kilometers (335 miles) per hour have been recorded in the subtropical jet stream, but because they are at an altitude of about 12 kilometers (39,000 feet), they have less influence on subtropical weather than the polar jet streams have on mid-latitude weather.

KEY POINTS IN THIS CHAPTER

In this portrait of the atmosphere, we have sought to describe its major characteristics—including its composition, dynamics, and general (global) circulation—as part of a larger description of the present-day climate system. Ultimately, however, the character of the atmosphere and, indeed, the state of Earth’s climate depend on the balance between incoming solar radiation and outgoing planetary radiation and how this balance varies over the planet. Therefore, the story of the atmosphere does not end here. Radiation balance is the conceptual basis for a quantitative description of climate change and places the phenomenon in a scientific context that we can systematically study. Because of this larger importance and its centrality to the subject, we investigate the concept of radiation balance separately in chapter 5.

To summarize what we have learned:

1. The atmosphere has been called the medium of climate because we live in it and because it connects the different parts of the climate system. It transports heat from the equator to the poles and from the surface to high altitudes; moves water evaporated from ocean to land, where it falls as rain or snow; and is an important pathway for carbon to flow among the various carbon reservoirs.
2. The main constituents of the atmosphere are N₂ (78.1 percent) and O₂ (20.9 percent). Other gases include Ar, H₂O, CO₂, CH₄, O₃, N₂O, and a host of “trace” gases. From a planetary perspective, Earth is unique because its atmosphere contains O₂, apparently due to the emergence of photosynthesis early in its history.
3. Because air is a compressible fluid, air pressure decreases exponentially with altitude. Consequently, nearly the entire mass of the atmosphere is in the troposphere. The density of surface air results from the mass of the overlying air (1.03×10^4 kg/m²).
4. The atmosphere is in continuous convective motion, which occurs on all scales from meters to thousands of kilometers.
5. Vertical motions of air result in adiabatic heating or cooling.
6. The amount of water vapor that the atmosphere can hold depends on temperature. This and adiabatic cooling account for the facts that the moisture content of the atmosphere decreases dramatically with altitude and that cold polar air is drier than warm tropical air.

7. The cooling of rising air masses causes water vapor to condense, transferring latent heat from the surface to altitude, and resulting in the formation of clouds and precipitation.

8. The thermal layering of the atmosphere also reflects how it is heated—the troposphere mainly from below and the stratosphere mainly from above—and how the greenhouse gases are distributed in it. Ozone accounts for the warmth of the stratosphere, but most of the greenhouse gases exist in the troposphere, in part simply because the density of air dwindles with altitude.

9. The lapse rate, or rate of temperature decrease in the troposphere with altitude, reflects the adiabatic decrease in density and the distribution of heat, water vapor, and other greenhouse gases through the troposphere by convective motions.

10. Essentially all water vapor is confined to the troposphere because the tropopause acts to trap it there. Without such a cold trap, Earth probably would have lost its water early in its history and become lifeless. Weather is also restricted mainly to the troposphere, with the overlying, relatively warm stratosphere representing, in effect, a stable lid on the troposphere.

11. About 90 percent of the O_3 in the atmosphere exists in the stratosphere, where ozone forms by photolysis. Ozone absorbs most of the incoming ultraviolet (UV) radiation, which would otherwise be lethal to life on the surface.

12. The general circulation of the atmosphere is caused by the uneven heating of the atmosphere and Earth's surface by incoming solar radiation, and by the Coriolis force. The circulation of the atmosphere acts to transport heat from the tropics to the polar regions and from the surface to the upper troposphere.

13. In the tropics and subtropics, global circulation is dominated by the Hadley cells, which account for the trade winds and desert belts at the 20° to 30° latitudes. The return flows of Hadley cell air meet to form the Intertropical Convergence Zone (ITCZ), an equatorial belt of low atmospheric pressure, high convection and rainfall, and high transfer of latent heat from the surface to the upper troposphere.

14. The mid-latitude zones of westerlies are known as the Ferrel cells. They originate from the breakup of Hadley circulation by Coriolis forces and are characterized by eddy circulation. Eddy circulation is responsible for nearly all of mid-latitude meridional transport of energy and moisture. Eddies exist as both quasi-stationary and transient waves.

15. The Ferrel and Polar cells meet at the polar front, where polar and tropical air masses interact to create numerous disturbances. The polar jet stream is the upper-tropospheric manifestation of the polar front.